

Curriculum Vitae

Tsung-Hsuan TSAI

Post-doctoral researcher

Department of Mathematics, Ashoka University, New Delhi

Website: <https://tsung-hsuan-tsai.github.io/>

Born: 13th April 1996

RESEARCH INTERESTS

- Geometric group theory: hyperbolic groups, random groups, small cancellation theory, van Kampen diagrams
- Topological dynamics: uniformly recurrent subgroups, topological groupoids

POSITIONS

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|-------------|--|
| 2025 - 2026 | Ashoka University, New Delhi, India
Postdoctoral researcher, with Suraj Krishna and Pritam Ghosh
Funded by Axis Bank Post-doctoral Programme |
| 2023 - 2025 | Institut Camille Jordan, Lyon, France
Postdoctoral researcher, with Nicolás Matte Bon
Funded by LabEx MiLyon |
| 2022 - 2023 | Career interruption (military service, Taiwan) |
| 2019 - 2022 | Institut de Recherche Mathématique Avancée, Strasbourg, France
Ph.D. in Mathematics, supervised by Thomas Delzant
Funded by CDSN (Contrat doctoral spécifique normalien) |

EDUCATION

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|-------------|--|
| 2019 - 2022 | Université de Strasbourg, France
Ph.D. in Mathematics, defended July 2022
Supervisor: Thomas Delzant
Thesis: <i>Phase transitions in random group: free subgroups and van Kampen 2-complexes</i> |
| 2016 - 2019 | École normale supérieure de Rennes, France
Normalien à titre étranger
M.Sc. in Mathematics at Université de Rennes 1, 2019
B.Sc. in Computer Science at Université de Rennes 1, 2018 |

B.Sc. in Mathematics at Université de Rennes 1, 2017

2014 - 2016 **Lycée Janson de Sailly, Paris, France**
Classe préparatoire aux grandes écoles, MPSI and MP

RESEARCH INTERNSHIPS

- 05 - 06 2019 **Institut de Recherche Mathématique Avancée, Strasbourg, France**
Research internship M2, supervised by **Thomas Delzant**
Master thesis: *Random groups: the binomial triangular model*
- 06 - 07 2018 **Università di Padova, Padova, Italy**
Research internship M1, supervised by **Remke Kloosterman**
Subject: *Fundamental groups of complements of singular algebraic curves*
- 05 - 06 2017 **Laboratoire de Mathématiques Jean-Leray, Nantes, France**
Research internship L3, supervised by **François Laudenbach**
Subject: *Algebraic topology and applications: Schreier Theorem and Poincaré paving*

TEACHING

- 2019 - 2022 **Université de Strasbourg, France**
Teaching assistant, 195 hrs
2019, 2020, 2021 autumn: **TD Algèbre S5**
Introduction to group theory, exercise sessions for 3rd year math students, 99 hrs.
2020, 2021, 2022 spring: **CI Math pour les sciences 2**
Basic linear algebra, lecturer for 1st year physics and chemistry students, 96 hrs.

PUBLICATIONS

- [1] T.-H. Tsai, *Density of random subsets and the intersection formula*.
In **Journal of Combinatorial Algebra**, volume 6, number 3/4, pp. 223–263 (2022).
DOI: [10.4171/JCA/63](https://doi.org/10.4171/JCA/63). ArXiv: [2104.09192](https://arxiv.org/abs/2104.09192).
- [2] T.-H. Tsai, *Freiheitssatz and phase transitions for the density model of random groups*.
In **Mathematische Zeitschrift**, 303, 65 (2023).
DOI: [10.1007/s00209-022-03186-2](https://doi.org/10.1007/s00209-022-03186-2). ArXiv: [2111.08958](https://arxiv.org/abs/2111.08958).
- [3] T.-H. Tsai, *Phase transition for the existence of van Kampen 2-complexes in random groups*.

In **Algebraic and Geometric Topology**, volume 24, issue 7, pp. 3897–3917 (2024).
DOI: [10.2140/agt.2024.24.3897](https://doi.org/10.2140/agt.2024.24.3897). ArXiv: [2210.08234](https://arxiv.org/abs/2210.08234).

- [4] T.-H. Tsai, *Parallel geodesics and minimal stable length of random groups*. In **International Journal of Algebra and Computation** Vol. 35, No. 08, pp. 1135–1158 (2025).
DOI: [10.1142/S0218196725500377](https://doi.org/10.1142/S0218196725500377). ArXiv: [2410.07859](https://arxiv.org/abs/2410.07859)

SELECTED TALKS

- 12.2025 **Séminaire virtuel francophone Groupes et Géométrie (online)**
Van Kampen diagrams and parallel geodesics in random groups
- 11.2025 **Mathematics Colloquium, Ashoka University, New Delhi, India**
Geometry in random groups
- 03.2025 **Analysis and geometry seminar, University of Bristol, UK**
Parallel geodesics and minimal stable lengths in random groups
- 03.2024 **Algebra and geometry seminar, LMNO, Caen, France**
Existence of van Kampen diagrams in random groups
- 02.2024 **GADT seminar, IMB, Dijon, France**
Existence of van Kampen diagrams in random groups
- 10.2023 **Groupe de travail "Action!", ENS Lyon, France**
Introduction to random groups II: The Freiheitssatz
Introduction to random groups I: The density model
- 12.2021 **ENS Group theory seminar, Paris, France (Online)**
Freiheitssatz for the density model of random groups
- 10.2020 **Séminaire Doctorant, IRMA, Strasbourg, France**
Most groups are hyperbolic

LANGUAGES

- English (fluent) • French (fluent) • Chinese (native)